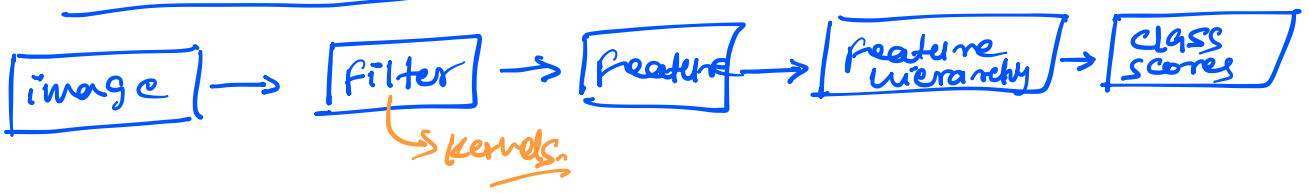
Introduction to Convolutional Neural Networks.

Keep the image structure. Learn what to look for.

Overview of CNN



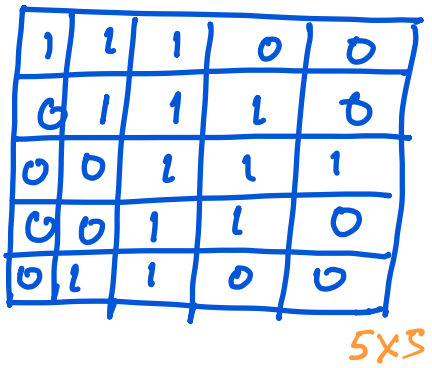
Why CNN:

1. Image remains as grid.
2. Locality: using nearby pixels rather than entire image to learn a feature.
3. Parameter sharing: use the small set of weights at every spatial position.
4. Spatial hierarchy: Early layers combine pixels to Edges. Later layers combines earlier features to detect shape and other parts.

Where CNNs are useful:

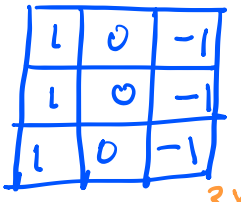
1. Images and videos
2. Grids → spectrograms, spatial maps
3. 1-D signals → 1D convolutions.
4. Graphs. → Graph conv. networks.

* Convolution



Image

X

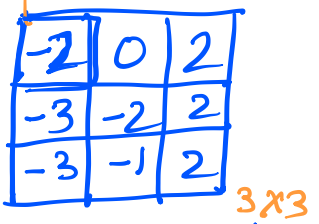


kernel

K

$1+0+(-1)+0+0+(-1)+0+0+0+(-1) = -2$

=



feature map

Y

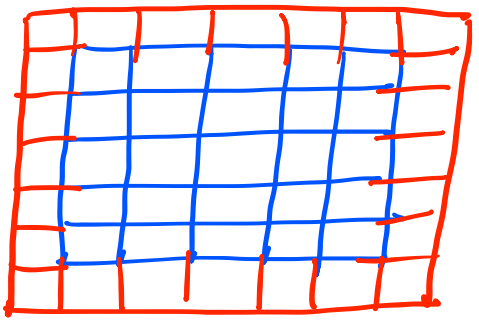
$$Y_{i,j} = \sum_{u=0}^{k_h-1} \sum_{v=0}^{k_w-1} X_{i+u,j+v} * k_{u,v}$$

In CNNs, the kernels are trainable parameters.

These are the weights.

$$K \leftarrow K - \eta \frac{\partial L}{\partial K}.$$

padding:



padding = 1

Before padding → 5X5

After " → 7X7

Kernel size → 3X3

Convolution → 5X5

4K → 3840 X2160

HD → 1920 X1080

→ 1280 X 720

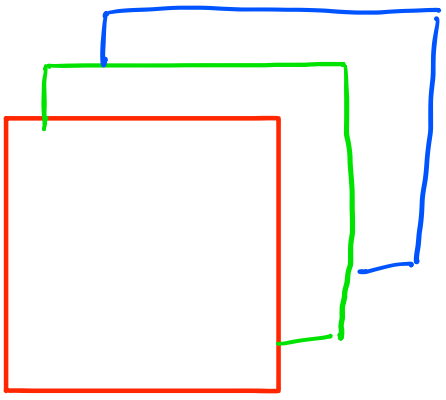
$$H_{out} = \left\lfloor \frac{H_{in} + 2P_h - K_h}{S_h} \right\rfloor + 1$$

$$W_{out} = \left\lfloor \frac{W_{in} + 2P_w - K_w}{S_w} \right\rfloor + 1$$

① Valid style convolution: $P = 0$, output shrinks.

② same-style convolution: $P = \frac{K-1}{2}$, with $S = 1$,
output size remains same

* Multichannel convolution:



X

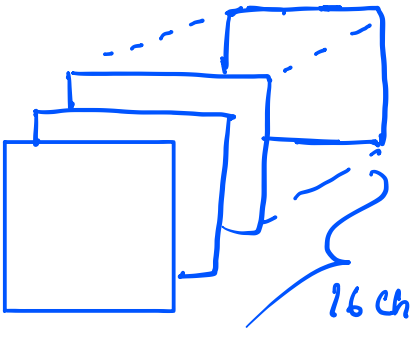
K_{Red}, K_{green}, K_{blue}
3x3 3x3 3x3

$* K \rightarrow$
(one kernel)



Y (1 channel)

K
(16 kernels) →



16 channels

weight calculation.

Convolution layer:

input → 3 channels

output → 16 channels

bias → false

kernel size = 3x3

hxwxC

MLP.

32x32x3 → 256

no. of weights = (32x32x3x256) + 256

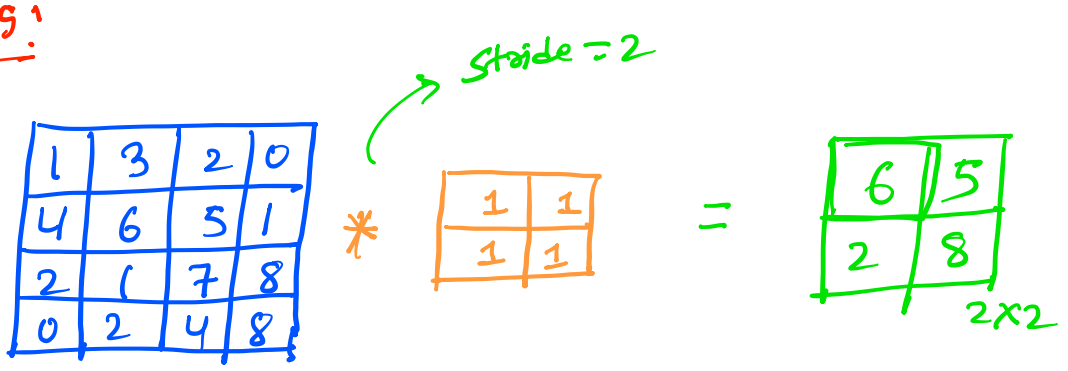
no. of channel in input.

hxw

Each of 16 filters 3x3x3 = 16x27 = 432

when bias = True, → (3x3x3)+1 = 16x(27+1) = 448.

* Pooling:



① Max-pooling → take the max

② Avg. pooling → take the avg.

Architecture:

